Underexposed Z63-domain synthetic example



Scene median: 0.25 DN (-3 EV from 2-DN source)

Common display compensation: +3 EV, white point: 1.5 DN

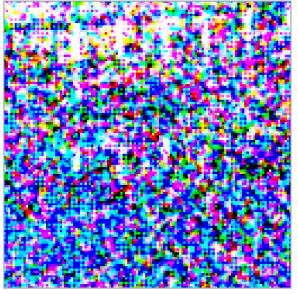
Frame 7, maximum injected-offset norm, fixed shadow crop below

Signal is reduced before shot/read noise and CFA offsets are added

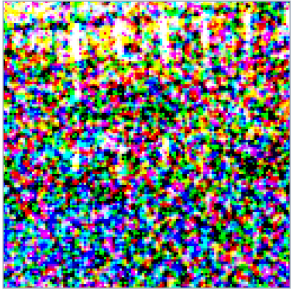
Frame 7: injected and estimated CFA offsets (DN)



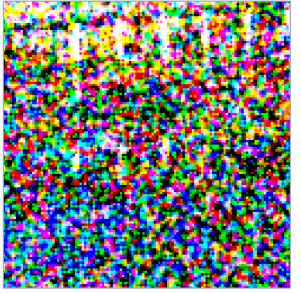
Uncorrected RAW



NAF offline T1



NAF sliding T7



Known-offset oracle



All panels share the same observation, bilinear demosaicing, +3-EV display recovery, and sRGB mapping.